Safe drinking water

Pollution

- Chemicals
- Waste
- microorganisms

effects

- Human
- Cholera
- diarrhea
- hepatitis
- environment
- aquatic death
- food chain disruption
- economic
- affects agriculture and Fisheries

prevention

- Treat industrial waste
- Reduce agriculture chemicals
- proper wage disposal
- Public awareness

Sources

- Rain water
- Surface water
- Ground water

Safe drinking water

- clear, Oderless, Colorless
- Pathogen-free, chemically safe

waste purification

- essential for Surface water
- Done in large scale treatment plants
- Remove pathogens and harmful substances

4 steps

- Intake
- Sedimentation "With alum"
- filtration "Rapid sand filter"
- Disinfection "chlorination" — essintial for Saftey

Alum forms folc — removes suspended matters

Filtration reduce bacteria to 5%

Chlorine:

- Dose: 1-1,2 PPM
- Residual: 0.1-0.2 ppm
- Contact time: 30 min

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Sinatry standards of Safe drinking water

Physical characteristics

clear, Oderless, colorless, Testeless, free from turbidity

chemicals characteristics

PH neutral or slightly alkaline "8"

TDS ≤ 500 PPM

Hardness ≤ 20 temporary: bicarbonate, permenati: Sulphates

indicators of organic pollution... organic matter ≤ 0,1 PPM

nitrites must not present

albuminoid ammonia ≤ 0,01 PPM

Nitratets ≤ 1 PPM Surface

oxygen absorbed ≤ 3-4 PPM

≤ 5 PPM ground

Biological characteristics

Total bacterial count ≤ 100/ml

Coliform must not present

no Pathogenic bacteria, viruses, or parasite

E.Coli fecal contamination

Strep. Faecalis recent

ColStridium Welchii old

Radiological characteristics

water must be free from it

Environmental health

Housing and health
good housing → prevent infections and improve mental health
poor housing → overcrowding, Infections, stress

environmental Sanitation
Contral of water, food, wastes
to prevent diseases

Environment Include all external
factors affecting human health

Health is a dynamic equilibrium
between man and environment

environmental hazards
Physical heat, noise, radiation
Biological microorganisms
chemical toxins, carcinogens, asphyxiants
Psychological Stress
Diet palated

environmental Sanitation
focus on Controlling harmful
environmental factors
objectives
improving quality of life
preventing disease
promotining hygiene
Controlling pests

environmental health studies and manage
environmental factors affecting health
Physical
Chemical
Biological
Social

Rural health

poor infrastructure → ↑ disease risk

Air Pollution

global effect
Acid rain, global worming
ozone depletion, radioactive disasters
Dioxine Pollution

contaminate of air
with harmful substances

Health effect
Short-term
irritation
infections
long-Term
COPD
Cancer
heart dis

Sources
Natural
human
vehicles is most important

indicators
Sulfur dioxide
carbon monoxide
nitrogen dioxide
oZone

natural cleaning
wind
sunlight
rain
Plants

High risk pregnancy

- identifyed by history examination, and investigations
- can developed at any stage of pregnaney

Causes:

maternal factors
age, parity, nutrition

Medical disease
DM, Hin, infections

Bad obastatic history

pregnancy complications
bleeding, pre-eclampsia, Twins

warning Signs:

vaginal bleeding
Severe headache
convulsions
Blurred vision
Severe abdominal Pain

• requier immediate medical attention

Intranatal care:

safe and clean delivery "5 cleans"
prevent infection
Manage complications
Care for newborn

Postnatal care:

Restore maternal health
Prevent complications
support breastfeeding
Provide family Planing

Major risks
Hemorrhage
infections

inter Pregnancy care

- Birth Spacing
- improve maternal health
- prepare for next pregnancy

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Maternal and child Health

compenant
- preconceptional Care
- Antenatal care
- Intranatal Care
- Postnatal Care
- inter pregnancy care
  - mother Care
  - Newborne care
  - Immunization
  - Family Planning

Key concepts
- continum of care
- prevention is the cornerstone

importance
- High risk Valerable group
- High morbidity and mortality
- most conditions are preventable

Mother and child are biologically and socially linked

Maternal condition directly affect fetal outcom

focus on preparing women before pregnancy
- medical screening
- Immunization
- Genetic Counselling
- lifesytlemodefication

Goals:
- Reduce risk factors
- improve maternal health
- prevent Gmplications

Antenatal Care:
care of women during pregnancy

Start early after conception

Components
- Regular check up
- investigations
- immunization
- Supplements

objectives
- maintain maternal health
- Detect high risk cases
- provide health education
- early diagnosis and treatment
- prepare for delivary and lactation

Key Interventions
- infection Screening
- Balanced diet, Physical activity
- folic acid supplementation
- Control Chloric disease
- emotional and fertility counselling

first visit:
Ideally 8 weeks "before 12–16"

includes:
- Pregnancy confirmation
- risk assessment
- Baseline investigation
- Start supplements
- health education

HIV

Risk factors
unprotected sexual activity
Needle sharing
maternal transmission

A RNA Retrovirus that attack CD4 T cells
weaking the Immune system

HIV is fragile outside the body

AIDS has high mortality rate
"80-90% within 3-5 years if untreated"

early infection is asymptomatic
but progression lead to Immunodeficincy

The advanced stage is AIDS, which
may take 10-15 years to developed

leads to Severe immuno damage
and susceptible to infections

long and variable
incubation period

not transmitted
by air, water,
food, and casual contact

epidemiology
Agent
(HIV-1) (HIV-2)
Reservior
Human
Transmission
Blood, Semen, vaginal secretions
and breast milk
Sexual transmission "most commen"
contaminated needle vertical transmission

- Hiv inters through skin branches and mucous membrane

- pateint are infectious from early infection until death
  —highst during window period and late Stages

Risk groups:
men who have sex with men
sex workers
intravenous drug users
Blood transfusion recipients

Clinical stages of HIV
- Primary infection
  occure 2-4 weeks after exposure
  Flue like symptoms
  highly infections
- symptomatic stage
  immune deterioration
  weight loss, diarrhea, Infections
- Asymptomatic stage
  last ~ 10 years
  No symptoms
  Antibody present
- AIDS
  Severe Immunosupprision
  CD4 < 200
  opportunistic infections
  and cancer

Diagnosis:
ELISA screening test
PCR Detect viral RNA early
CD4 Count indicator of disease progression

☆ window Period: false negative test despite Infection

Hiv prevention Stratiges

.IEC "information, education, comunication"

promote awareness, safe sex, and behavior change

.Blood borne prevention

screening of Blood
sterilization of instruments
Avoid sharing needs

.mother to child prevention

Antiretroviral drugs for mother and baby
Cesarean delivery
Avoid breastfeeding

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.post exposure prophlaxis

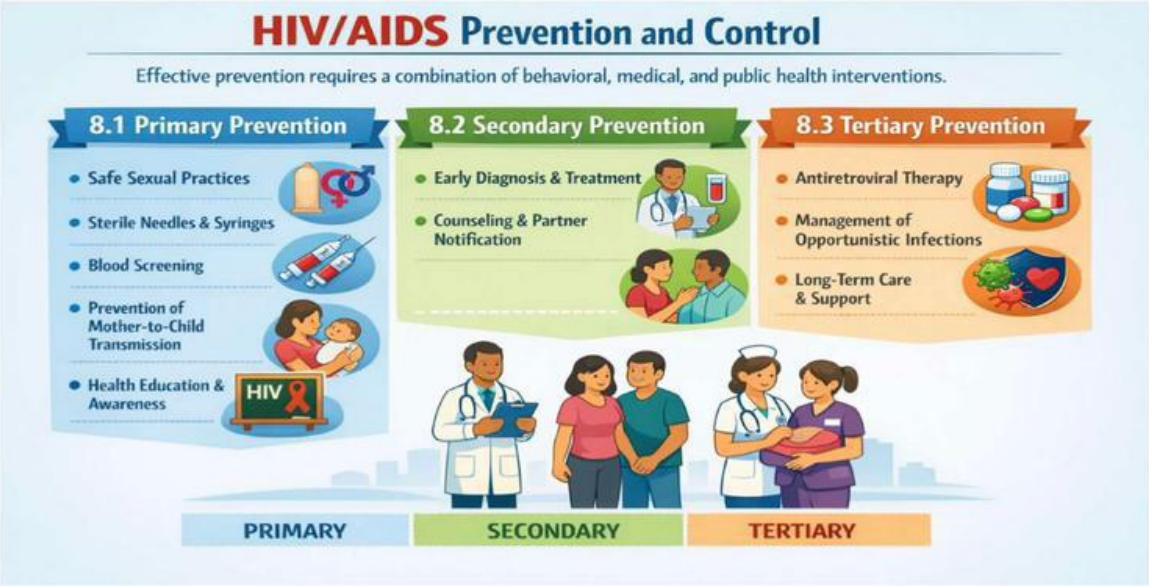
Started within 72 hours
continued for 28 days

.preexposure prophlaxis

Daily medications for high risk Individuals

.Antiretroviral therapy "ART"

Reduce viral load
improves immunity and quality of life



Substitution of resources

Replace one resource with another when needed

Ensures continuity of work

functions determine structure

organizational structure is based on defined functions

Delegation of authority

Assign authority and responsibility to others

Management by exception

focus only on deviations or problems

Temporary transfere of authority

To Perform tasks

improve dicision speed and efficiency

Decision should be made close to the action site

Acceptance

Assignment of tasks

Process of delegation

Accountability

Transfere of authority

principles

Define clear results. Select qualified Persons. Set authority limits. Provide resources. monitor and follow up

importance of management

optimal use of resources

Motivation of workers

Team Spirit

stability and growth

Benifits

increased efficiency

Reduce turnover

Higher productivity

smooth functioning

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Management

Decision making
core of management
involves Identifying
problems, evaluating
altranatives, and
selectingthe best
Solution

Resources
manpower
materials
money

It's the process of getting
things done through people by
using recources efficiently
and making informed decisions

involve setting clear objectives
and ensuring their achiument

functions

Planning

- setting goals and decide How to achive them
- involves forecasting, decision making and resource allocation

organizing

- Arranging tasks, people and resources
- Define roles, responsibilities, and authority
- improves efficiency and avoids duplication

leading

- guiding and motivating people to achieve goals
- focuses on team work, communication, and leadership

Controlling

monitoring performance and ensuring goals are met

Steps:

Set Standards, measure performance

. compare results . Take corrective actions

Smart objectives
specific
measurable
Achievable
Relevant
Time-bound

Goals Broad aims
objectives specific, measurable steps
out comes measurable results

principles of management

Managment by objectives

Set clear measurable goals
compare results with targets
dentify gaps and improve performanc

learning from experience

Analyze differences between expected
and actual results
improve future decisions

Division of labor

work is divided based on specialization
increase efficiency

coordination of activities

Align all activites toward a common goal
Promotes teamworke